

agentic-memory

Long-Context Benchmark Report

Benchmark evidence for >300k-token conversations comparing short-context prompting, full transcript stuffing, and agentic-memory retrieval.

Pass Rate

57.0%

memory_enabled

Prompt Tokens

349.4

avg per answer

Cost vs Full

242.5x cheaper

memory_enabled

Headline

On >300k-token conversations, agentic-memory retrieval outperformed both short-context and full-context baselines while using dramatically fewer prompt tokens than transcript stuffing.

Key Findings

- ? memory_enabled achieved a 171/300 pass rate (57.0%), versus 72/300 (24.0%) for full_context and 4/300 (1.3%) for short_context.
- ? memory_enabled used 349.4 prompt tokens on average, compared with 160543.9 for full_context.
- ? memory_enabled was 58.4% faster on average than full_context.
- ? full_context cost about 242.5x more per example than memory_enabled.

Mode Comparison

Measured on 12 beyond-250k conversations with Gemini 2.5 Flash.

Mode	Pass	Judge	Latency	Prompt Tokens	Cost
memory_enabled	171/300 (57.0%)	0.73	1045 ms	349.4	\$0.000066
full_context	72/300 (24.0%)	0.32	2512 ms	160543.9	\$0.016114
short_context	4/300 (1.3%)	0.11	980 ms	1718.4	\$0.000196

Business Impact

- ? Accuracy lift over full_context: 33.0 percentage points.
- ? Latency reduction versus full_context: 58.4%.
- ? Prompt-token reduction versus full_context: 99.8%.
- ? Estimated cost reduction versus full_context: 99.6%.

Interpretation

On transcripts that grow beyond a practical prompt budget, brute-force context stuffing becomes expensive and still leaves quality on the table. Retrieval through agentic-memory preserves the relevant profile facts, prior events, and successful procedures while keeping prompts compact enough to stay economical.

Recommended Next Step

Tune the memory-mode answer scaffold so Gemini is required to explicitly cover every gold constraint that appears in retrieved memory. That should push the pass rate higher without giving up the cost and latency advantage.

Sample Size Note

This 12-example run is a pilot. For a public-facing directional benchmark, the next target should be at least 100 total examples, ideally with 20 or more per scenario family. For tighter 95% pass-rate intervals near plus or minus 5 percentage points, target roughly 385 to 400 total examples.